- *Cloud integration*—AWS IoT Core provides managed MQTT infrastructure along with integration to other cloud services for storage, processing, and analysis. The LoRaWAN Network Server (LNS) and Configuration and Update Server (CUPS) handle the complexity of managing LoRaWAN devices while presenting a simple MQTT interface to applications.

3.7.4 The processing layer: message decoding

Initial processing of sensor data is performed in the message decoder function where device-specific data formats from different types of sensors are converted to a common data format before being routed for additional processing and storage.

3.7.5 Data flow walkthrough

To understand how this architecture works in practice, let's trace a sensor reading from physical measurement to digital twin application:

1. *Physical measurement*. As a garden tap is turned on and water flows through a flowmeter, spinning the blades on the impeller that have magnets attached to them. The rotating magnets create a changing magnetic field that is detected by the Hall effect sensor. This sensor outputs a signal, typically a frequency, proportional to the flow rate.
2. *Signal processing*. The sensor's electronics convert soil conductivity to a digital moisture percentage, apply calibration, and format the reading with timestamp and device identifier.
3. *Communication*. The sensor transmits a compact LoRaWAN packet containing the moisture reading, device ID, and metadata.
4. *Gateway processing*. The LoRaWAN gateway receives the radio signal, validates the message, and forwards it over the internet to AWS IoT Core.
5. *Network server processing*. The AWS LoRaWAN Network Server authenticates the device, decrypts the message, and publishes the data to an MQTT topic following the naming convention `home/water/garden/flow_meter_1`.